

Syed Mujahid | Aero Design Portfolio

About Me

I am a Design Engineer specializing in aerospace conceptual design, advanced CAD modeling, mechanical architecture, and prototyping. My focus is on developing practical, manufacturable solutions, taking ideas from conceptual sketches through structural validation (FEA) and into production-ready prototypes.

Beyond my core technical expertise, I have strong experience in cross-functional coordination, Project collaboration, and technical documentation for advanced aircraft development programs. I bring a solid aeronautical engineering background to ensure seamless integration and communication across all phases of a project.

My work spans complete aircraft conceptual design, mission analysis, and system-level mechanical development for projects such as:

- **Heavy-Lift Cargo VTOLs (40kg MTOW) featuring hybrid DfAM structures**
- **19-passenger amphibious aircraft (ME-1A)**
- **eVTOL air ambulances and supersonic UCAV concepts**

I follow structured methodologies inspired by Mohammad H. Sadraey's *"Aircraft Design: A Systems Engineering Approach"*, applying a precise balance of aerodynamic principles, structural integrity, and practical manufacturing execution to every project.

Table of Contents

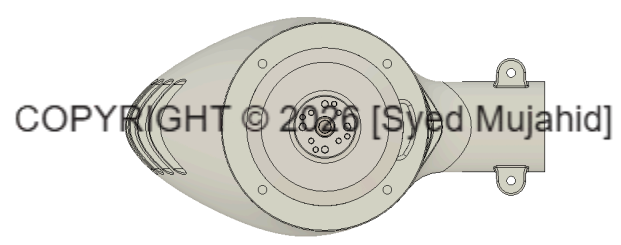
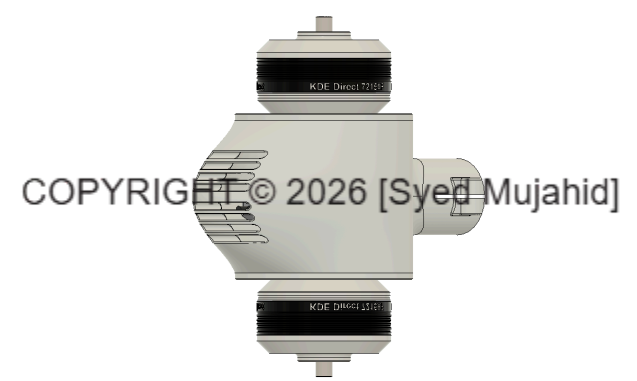
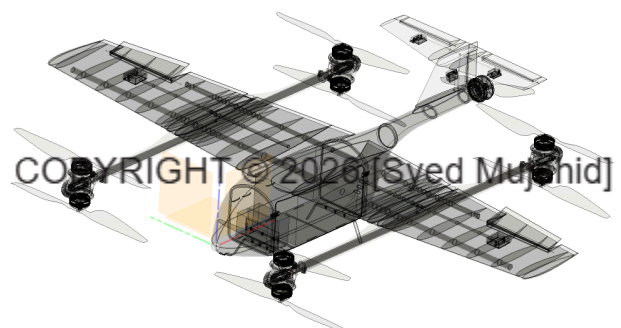
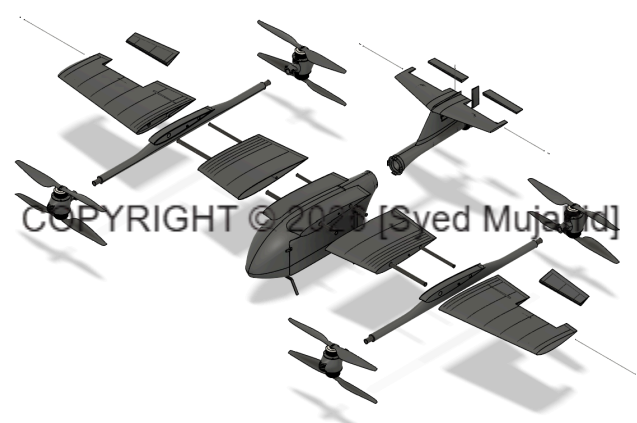
1. Heavy-Lift Modular Cargo VTOL UAV (40kg MTOW)
2. Amphibious Aircraft (ME-1A)
3. Air Ambulance eVTOL
4. Supersonic Unmanned Combat Aerial Vehicle (UCAV)
5. Passenger eVTOL – 3D Concept Design
6. Modified Concorde Aircraft – Freelance Project

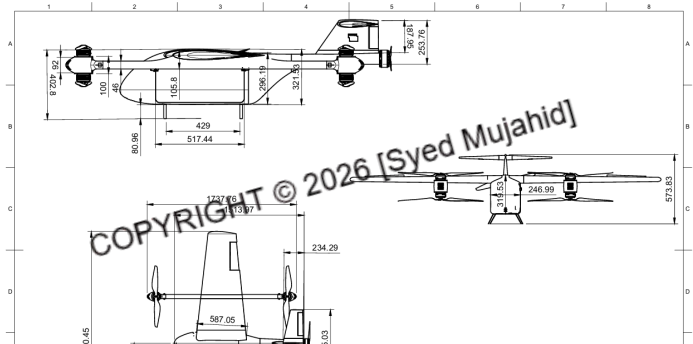
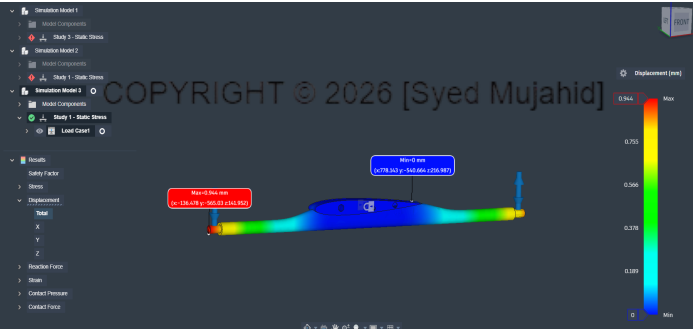
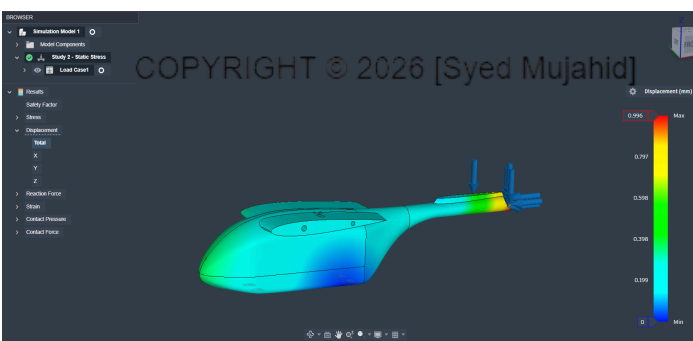
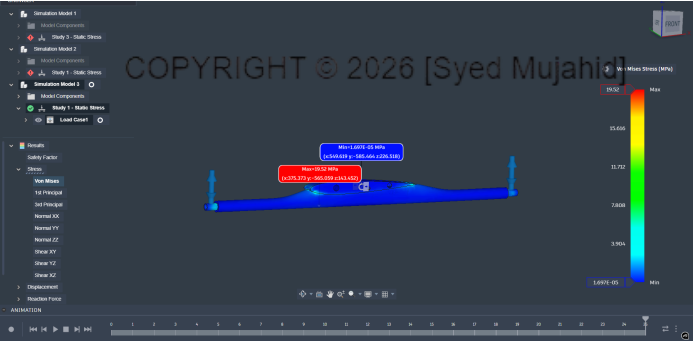
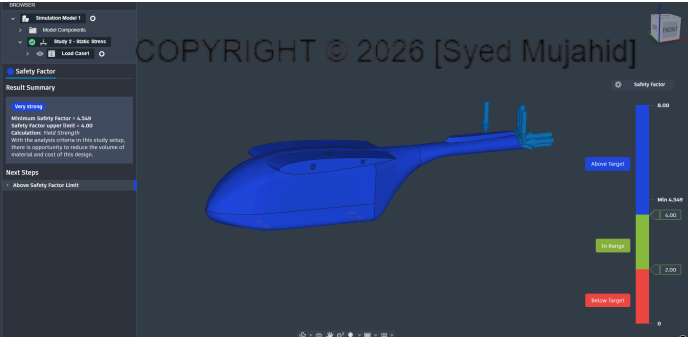
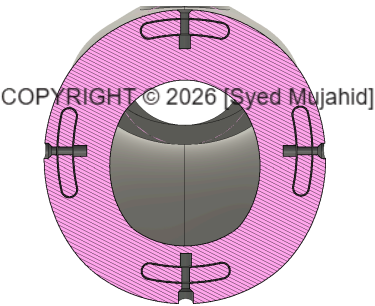
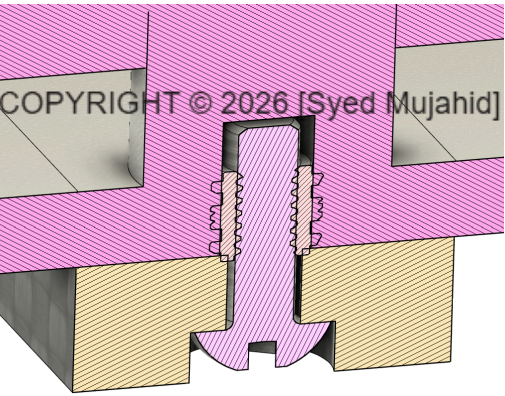
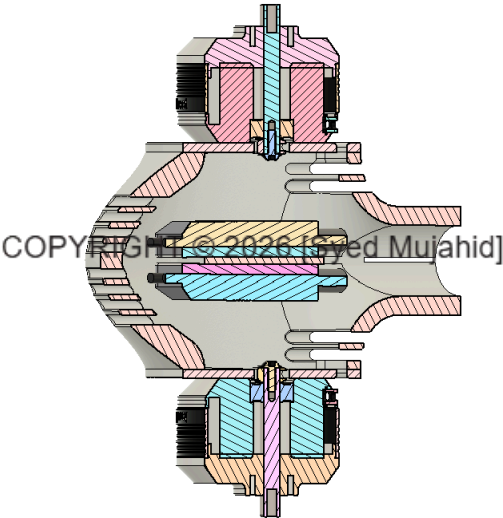
1. Heavy-Lift Modular Cargo VTOL UAV (40kg MTOW)

Role: Design Engineer

Tools Used: Fusion 360, Openvsp

Description: Engineered an end-to-end, manufacturing-ready 40 kg MTOW fixed-wing VTOL UAV in Fusion 360, utilizing Design for Additive Manufacturing (DfAM) principles to develop a modular, FEA-validated, 3D-printable hybrid airframe with fully integrated propulsion, thermal, and avionics systems.





2. Amphibious Aircraft (ME-1A)

Role: Aircraft Design Engineer

Tools Used: Fusion 360, Blender, Openvsp

Description: Conceptual Designed & Developed CAD models and design configurations for a 19-passenger amphibious aircraft. Created multiple mission variants, including passenger, cargo, and military special operations.

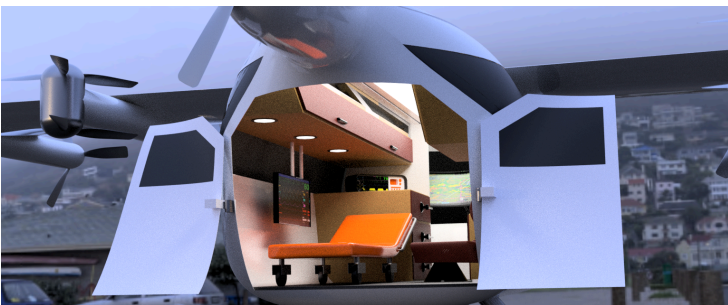


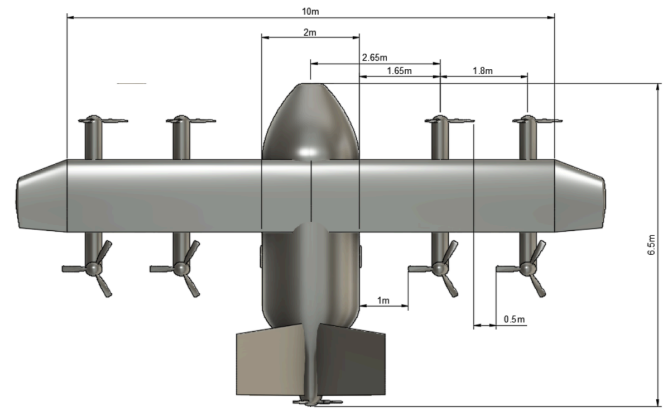
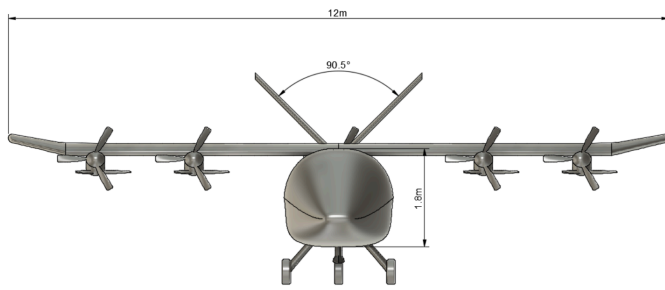
3. Air Ambulance eVTOL

Role: Lead Designer in Nacdec Competition

Tools Used: Fusion 360, Openvsp

Description: Led a team of 5 in conceptualizing and modeling an Air Ambulance, cabin layout optimized for patient transfer, medical equipment integration, and crew safety. Presented my work in the advanced air mobility institute.



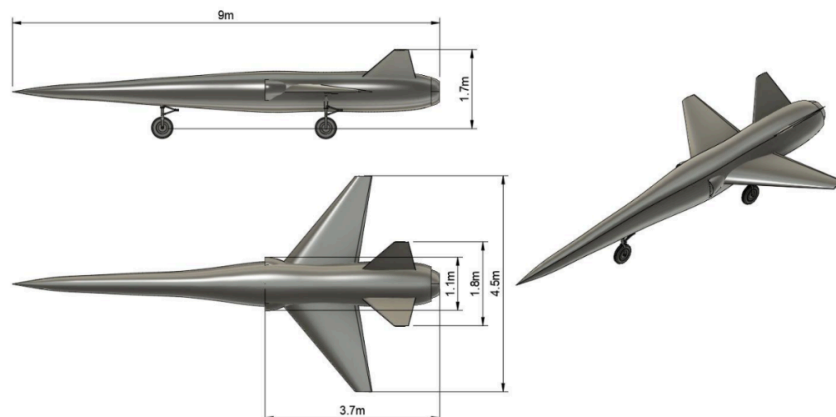
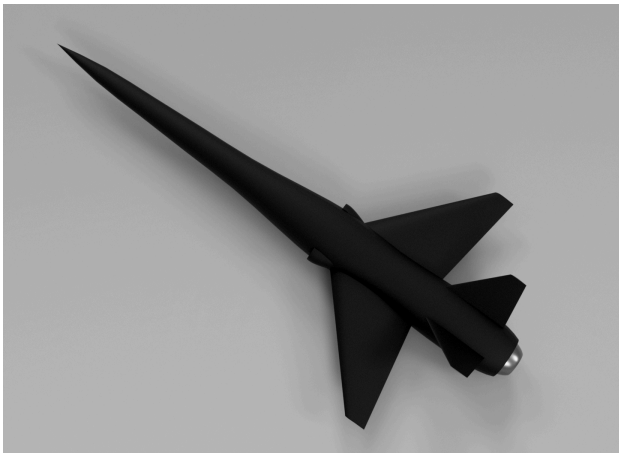


4. Supersonic Unmanned Combat Aerial Vehicle (UCAV)

Role: Concept Designer

Tools Used: Fusion 360, Mission Analysis Tools, Openvsp

Description: Worked on the conceptual design of a supersonic unmanned combat aerial vehicle optimized for long-range strike missions, survivability, and high-speed performance. The design study focused on aerodynamic efficiency, mission requirements, and system-level integration.

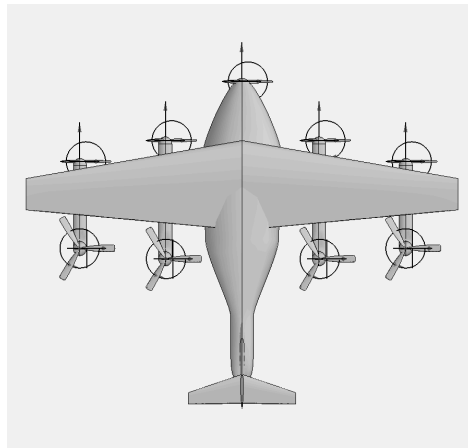
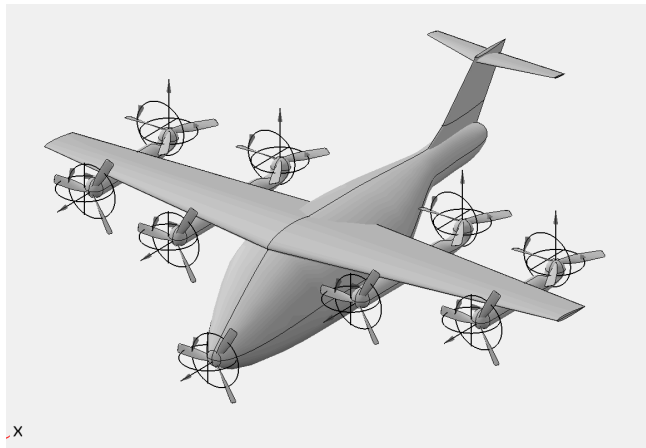
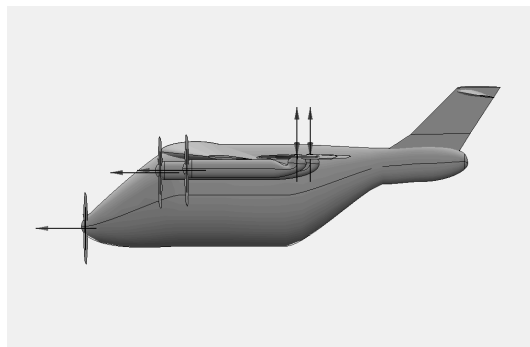
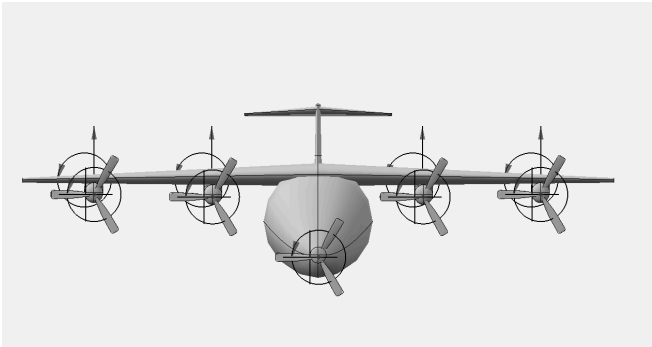
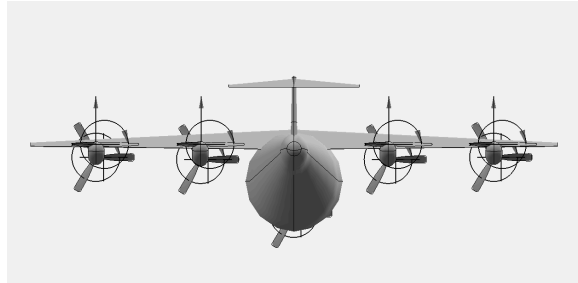


5. Passenger eVTOL – 3D Concept Design

Role: Lead Concept Designer

Tools Used: Fusion 360, Openvsp

Description: Developed a passenger eVTOL aircraft concept focused on urban and regional air mobility.



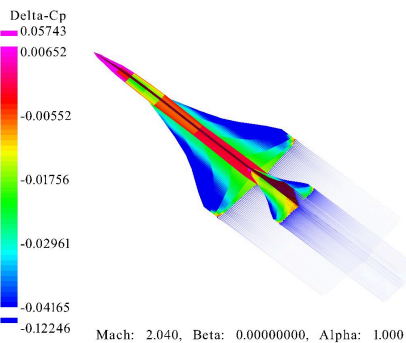
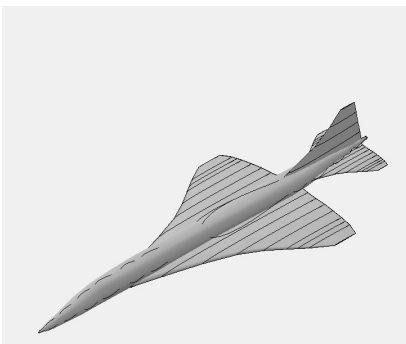
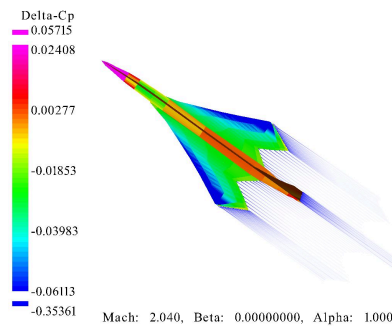
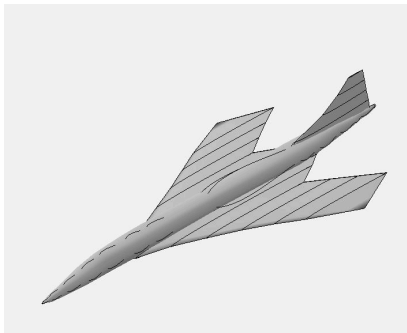
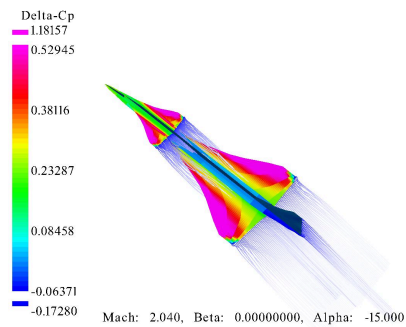
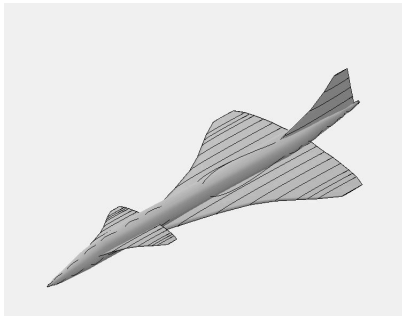
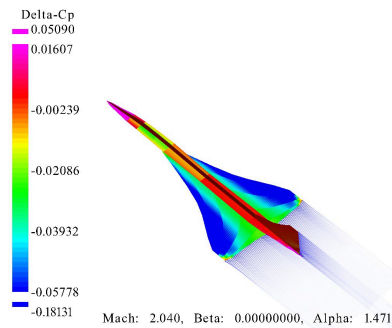
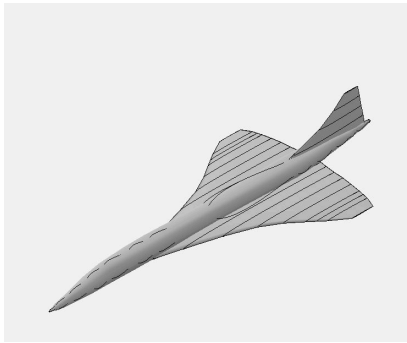
6. Modified Concorde Aircraft – Freelance Project

Role: Freelance Designer

Client: Private Client (England)

Tools Used: Openvsp

Description: Successfully completed a freelance project designing a modified Concorde aircraft for an international client. The work involved developing multiple conceptual configurations, conducting aerodynamic analyses, and preparing deliverables that combined both technical depth and design clarity.



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